

Merge Sort

Unsorted array

5	4	1	8	2
left 1		right 1		

Call merge sort to left 1

5	4
left 2	right 2

Call merge sort to left 2

5

→ If it is a single
element arrays are always sorted.

1 element → return, left 2 is sorted.

Call merge sort to right 2

4

1 element → return, right 2 is sorted

Since left 2, right 2 is sorted, we
have to call the merge algorithm
for left 2, right 2 in a sorted
manner.

4	5
---	---

→ left 1

Call merge sort to right 1

1	3	2
left 3		right 3

Call merge sort to left 3

1

Call merge sort to right 3

3	2
left 4	right 4

Call merge sort to left 4

3

Call merge sort to right 4

2

Then sort it as,

1	2	3
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Merge left 1 and right 1

1	2	3	4	5
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 → sorted array

Pseudocode

Algorithm : merge sort

Input : Unsorted array, A

Output : Sorted array.

Let's write the base case

$n \leftarrow A.length;$

if ($n == 1$)

{

return A;

}

// create left and right subarrays

left = mergesort(left);

right = mergesort(right);

merged array = merge(left, right)

return merged array;

Time Complexity

$$T(n) = \begin{cases} 1 & n = 1 \\ T(\frac{n}{2}) + T(\frac{n}{2}) + cn & \end{cases}$$

$$T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + cn$$

$$T(n) = 2T(\frac{n}{2}) + cn \quad \text{--- ①}$$

$$T(\frac{n}{2}) = 2T(\frac{n}{4}) + c(\frac{n}{2}) \quad \text{--- ②}$$

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + c\left(\frac{n}{4}\right)$$

$$T(n) = 2 \cdot 2T\left(\frac{n}{4}\right) + c\left(\frac{n}{2}\right) + c_n$$

$$= 4T\left(\frac{n}{4}\right) + \cancel{2 \cdot c \cdot \frac{n}{2}} + c_n$$

$$= 4T\left(\frac{n}{4}\right) + 2c_n$$

$$T(n) = 4 \left[2T\left(\frac{n}{8}\right) + c \cdot \frac{n}{4} \right] + 2c_n$$

$$= 8T\left(\frac{n}{8}\right) + \cancel{4 \cdot c \cdot \frac{n}{4}} + 2c_n$$

$$= 8T\left(\frac{n}{8}\right) + 3c_n$$

Generalize

$$= 2^i T\left(\frac{n}{2^i}\right) + i c_n$$

$$\frac{n}{2^i} = 1$$

$$n = 2^i$$

Apply log on both sides

$$\log n = \log 2^i$$

$$\boxed{\log n = i}$$

$$\begin{aligned} & n T(1) + \log n \cdot c \cdot n \\ & n \cdot 1 + c \cdot n \cdot \log n \\ & n + c \cdot n \cdot \log n \end{aligned}$$

$$\boxed{O(n \log n)}$$